UNIVERSITY OF ICELAND

Wind Speed and Rural Traffic Accidents in Iceland

A Frequency-Adjusted Analysis, 2007–2025

Ásgeir Snær Magnússon

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M.Sc. thesis
in Software Engineering

FACULTY OF INDUSTRIAL ENGINEERING, MECHANICAL ENGINEERING AND COMPUTER SCIENCE

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30 ECTS thesis submitted in partial fulfillment of a
Master of Science degree in Software Engineering

Supervisors
Kristján Jónasson
Guðmundur Freyr Úlfarsson

Faculty of Industrial Engineering, Mechanical Engineering and Computer Science
School of Engineering and Natural Sciences
University of Iceland
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Faculty of Industrial Engineering, Mechanical Engineering and Computer Science
School of Engineering and Natural Sciences
University of Iceland
Dunhagi 5
107, Reykjavik, Iceland
Telephone: 525 4000

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Abstract

This thesis examines whether strong mean wind speeds are disproportionately common when rural injury accidents occur in Iceland. Accident records, cleaned 10-minute weather observations, urban boundaries, annual road-section traffic statistics, and daily counter data are combined in a documented analysis. The primary analysis includes 6,192 rural injury accidents from 2007–2025 with valid wind observations within 20 km. Within each weather station and meteorological season, observed accidents in each mean-wind interval are compared with the number expected from the local background frequency of that interval, pooled across 2007–2025. Confidence intervals are obtained from 5,000 bootstrap samples clustered by weather station.

The observed/expected ratio is close to one across most low and moderate wind intervals. At mean wind speeds of 20–25 m/s, 61 accidents were observed compared with 29.0 expected, giving a ratio of 2.11 (95% station-clustered bootstrap interval 1.54–2.73). Wind gust at the matched accident time is a secondary wind measure. The result describes relative accident occurrence, not a causal effect or accident risk per vehicle. Complete hourly traffic exposure is unavailable at the required spatial and temporal resolution for the full study period. The thesis documents data cleaning, matching, coverage and statistical analysis separately.

Útdráttur

Ritgerðin rannsakar hvort hár meðalvindhraði sé óvenju algengur þegar umferðarslys með meiðslum verða í dreifbýli á Íslandi. Hún sameinar slysaskrá, hreinsuð tíu mínútna veðurgögn, þéttbýlismörk, árleg umferðargögn fyrir vegkafla og daglegar umferðartalningar. Aðalgreiningin nær til 6.192 slysa með meiðslum á árunum 2007–2025 sem tengjast gildri vindmælingu innan 20 km. Fyrir hverja veðurstöð og hverja árstíð er fjöldi slysa í vindbili borinn saman við þann fjölda sem vænta mætti miðað við staðbundna vindtíðni sem er safnað yfir öll ár rannsóknartímabilsins.

Við meðalvind upp á 20–25 m/s urðu 61 slys, en 29,0 voru vænt. Hlutfall séðra og væntra slysa var því 2,11 (95% bil 1,54–2,73). Vindhviður eru skoðaðar sem aukagreining. Niðurstaðan lýsir hlutfallslegri tíðni slysa, en ekki orsakaáhrifum eða áhættu á hvert ökutæki. Umferðargögn eru ekki tiltæk með nægilega nákvæmri staðsetningu og tímasetningu fyrir allt rannsóknartímabilið.

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Abbreviations

ADU	Average annual daily traffic.
SDU	Summer daily traffic, June–September.
VDU	Winter daily traffic, December–March.
VHDU	Spring/autumn daily traffic derived from ADU, SDU and VDU.
O/E	Observed accidents divided by expected accidents.

1. Introduction

Strong wind can affect vehicle stability, lane keeping, driver behaviour, and the decision to travel. Iceland is a useful setting for studying this question because wind conditions vary strongly across locations and time. However, accident counts alone are not accident risk. A severe weather event can raise the risk faced by each vehicle while simultaneously reducing the number of vehicles on the road. Weather frequency and traffic exposure must therefore be treated as separate parts of the problem [Andrey et al.(2003), Lord and Mannering(2010), Washington et al.(2020)].

The primary analysis focuses on mean wind speed and rural accidents involving injury. Matched-time wind gust and traffic results are reported separately.

1.1. Previous Research

Crosswind can produce overturning, lateral sideslip, or excessive yaw. The critical condition depends on vehicle geometry and mass, vehicle speed, road friction, wind direction, and driver response rather than wind speed alone [Baker(1986), Baker(1994)]. High-sided vehicles are particularly exposed, but the physical mechanisms also provide a reason to examine loss-of-control accidents more generally. These engineering studies establish physical plausibility and vehicle-specific failure mechanisms; they do not by themselves establish a population-level association between wind and accident occurrence.

This problem is directly relevant in Iceland, where terrain can accelerate and redirect near-surface wind over short distances. An Icelandic reliability study modelled accident probability as the combined outcome of wind velocity and direction, road camber and friction, and vehicle speed [Snæbjórnsson et al.(2007)]. That work explains why a weather station is an imperfect proxy for conditions at a specific road location and why spatial matching sensitivity is necessary in the present study. Mobile road- measurement research similarly shows that roadside terrain creates marked spatial variation that a sparse fixed-station network cannot fully resolve [Pu et al.(2018)].

The operational relevance is not hypothetical. Vegagerðin has developed a vehicle-wind model to support decisions about hazardous conditions and road closures, while also noting the limited experimental evidence available for passenger cars [Hrafnkelsdóttir(2020)]. The present thesis does not test that operational model; it examines population-level accident occurrence using a separate data design.

Empirical work has compared the share of accidents occurring in high wind with the share of time that high wind was observed, an approach closely related to the observed/expected comparison used here [Edwards(1994)]. Studies linking crashes to nearby weather stations likewise show that station observations can be informative while retaining exposure error [Young and Liesman(2007)]. More broadly, reviews find that road-safety evidence for wind is less consistent than for precipitation and emphasise the need to distinguish weather occurrence from traffic exposure [Theofilatos and Yannis(2014)].

Weather can change the denominator as well as the hazard. Research using automatic traffic recorders shows that extreme weather variables, including wind, can reduce or redistribute hourly traffic volume [Sathiaraj et al.(2018)]. Accident counts during strong wind can therefore remain low even if risk to each travelling vehicle increases. High-resolution crash studies further demonstrate the value of hourly weather and time-stratified comparisons rather than daily or monthly averages [Xing et al.(2019)]. The present thesis consequently separates background wind frequency, estimated vehicle-kilometres, and observed daily traffic instead of treating them as one interchangeable exposure measure. Studies with real-time traffic, weather, and road geometry illustrate what a more complete joint exposure design can achieve when those data are available [Ahmed et al.(2012)].

1.2. Objective and Research Question

The objective is to determine whether rural injury accidents are more common under strong mean wind than would be expected from the local frequency of those conditions. The primary research question is:

> How does the relative occurrence of rural injury accidents in Iceland vary across mean wind-speed intervals after standardisation for wind frequency at the matched weather station and season?

Wind gust from the observation matched to the accident time is examined with the same method as a secondary exposure. The temperature and daylight conditions at accident times are described as exploratory context. Temperature is also compared with its local seasonal frequency, but it does not replace mean wind as the primary exposure. The available annual and daily traffic data are then used to assess whether the main pattern remains plausible after accounting, as far as the data allow, for changes in traffic exposure. These are supporting analyses rather than additional primary research questions.

The primary result estimates relative accident occurrence. It does not estimate a causal effect or a national accident rate per vehicle-kilometre.

The analysis records weather-cleaning rules, keeps unmatched accidents in the matching data, reports spatial coverage, calculates local background wind frequency, and estimates uncertainty by weather-station-clustered bootstrap. The figures and tables can be regenerated from documented scripts [Peng(2011)].

2. Data Sources and Analysis Files

2.1. Source Websites and Downloaded Data

Table 2.1 lists the downloaded source data. The identifiers in the first column are used in Table 2.3. The source documentation is cited in the bibliography [Samgöngustofa(2026), Veðurstofa Íslands(2026), Vegagerðin(2026), Hagstofa Íslands(2026)].

Table 2.1: Downloaded source data. The primary study period is 2007–2025.

ID	Source website	Downloaded data	Records and period
ACC	Samgöngustofa	Accident events and road links	126,607 events and 126,610 road links, 2007–2025
IMO	Icelandic Meteorological Office	Ten-minute weather observations and station metadata	226,580,952 observations and 771 station-history records, 2007–2025
VGD-A, VGD-D, VGD-R	Vegagerðin	Annual road traffic, daily counter reports, counter locations, and road geometry	33,757 section-years, 2000–2025; 1,033,659 counter channel-days, 2019–2024
SI	Statistics Iceland	Urban-area polygons	97 polygons, 2020–2024 publication

The 2025 accident, injury, vehicle, road-link, annual-traffic, and weather records pass through the same preparation rules as the earlier years and are included in the primary analysis. Daily counter reports currently end in 2024.

Table 2.2 shows the first chronological records in the accident-event delivery. The source identifier is named `nid` or `NID` in the supplied files and is renamed `id` at import. Its values are source record keys, not row numbers, so chronological records are not expected to be numbered 1, 2, 3, and so forth. The values are retained unchanged because they link the accident, road-link, and vehicle files. The analysis also retains the original injury and accident-type codes.

Table 2.2: First chronological records in the accident-event delivery. Location text is shortened only for display.

id	Date	Time	x	y	meidsli	tegohapps	stadsetn
3166374	01.01.2007	00:38:00	361264	405675	3	121	Á Sæbraut, skammt norðan við Miklubrautarbrúna
3166372	01.01.2007	00:44:00	358064	406363	4	920	Við gatnamót á Bústaðavegi, Select
3166394	01.01.2007	01:14:00	359475	405515	4	142	Á Bústaðavegi, LHS, gegnt
3166406	01.01.2007	01:27:00	355877	400981	4	270	Við Miðvang, hús nr. 123
3166596	01.01.2007	04:42:00	356779	408494	4	920	Við Garðastræti 6
3166700	01.01.2007	06:31:00	356995	407945	4	710	Við Skálholtsstíg 2
3166730	01.01.2007	06:58:00	358845	405767	4	510	Gatnamót Bústaðavegar og Brúarinnar
3166883	01.01.2007	13:44:00	277239	496763	4	920	Gatnamót Ólafsbrautar og Ennisbrautar
3166887	01.01.2007	13:48:00	401235	384366	3	12	S-vegur, Fossnes 14
3166897	01.01.2007	14:17:00	484011	561541	4	21	Sfj.-vegur, Syðsta-Grund

Table 2.3 records the exact file transformations used to create the analysis data. Dataset contents and purposes have already been defined in Table 2.1 and the preceding text. To keep the table compact, script paths are relative to `src/` and data paths are relative to `data/`. Collections of unchanged annual workbooks and daily reports are shown with a file pattern. Within a cell, each shared directory is printed once and the following filenames belong to that directory.

Ordinary analysis begins only after the compact CSV layer has been exported. Table 2.4 lists the retained analysis-to-result steps; none reads from `raw/` or `processed/`. Input paths beginning with `analysis/` are relative to `data/`; result paths are relative to `reports/`.

2.2. Accident Data

The accident source covers 2007–2025 [Samgöngustofa(2026)]. The study population comprises 6,414 rural accidents involving injury. Accident severity is based on the source field `meidsli`: code 1 denotes a fatal accident, code 2 a serious accident, code 3 a minor-injury accident, and code 4 an accident without injury. The primary outcome is therefore `meidsli` ≤ 3. The serious-or-fatal comparison uses `meidsli` ≤ 2. The source data also contain 17,591 rural damage-only accidents; these are outside the injury outcome.

Rural status is derived from accident coordinates and Statistics Iceland urban-locality polygons [Hagstofa Íslands(2026)]. Points inside an urban polygon are classified as

Table 2.3: Exact inputs and outputs of the data-preparation pipeline

Script	Input file(s)	Output file(s)	Description
accidents/build.py	*raw/accidents/* accidents_*.txt vehicles_*.txt road_links_2007_2025.txt urban_boundaries_2020_2024.geojson	*processed/accidents/* all.csv	Joins the accident sources and retains the fields needed for selection and matching.
weather/clean.py	*raw/weather/* weather_10min_raw.parquet	*processed/weather/* weather.parquet	Applies the fixed ten-minute weather-quality rules.
weather/frequency.py	*processed/weather/* weather.parquet	*processed/weather/* frequency.csv traffic_frequency.csv	Counts pooled station-season weather exposure for O/E and year-specific mean-wind exposure for traffic models.
traffic/annual.py	*raw/traffic/annual/* *.xls*	*processed/traffic/* annual.csv	Standardises road section, length, ADU, SDU and VDU.
accidents/match_weather.py	*processed/accidents/* all.csv *processed/weather/* weather.parquet *raw/weather/* stations.csv	*processed/accidents/* rural_injury.csv	Independently matches wind and temperature within 20 km and five minutes; retains wind matches to 30 km for sensitivity.
accidents/case_control.py	*processed/accidents/* rural_injury.csv *processed/weather/* weather.parquet	*processed/accidents/* case_control.csv	Selects same-hour, same-weekday control times in each accident month for wind and temperature.
traffic/build_road_period.py	*processed/traffic/* annual.csv *processed/accidents/* rural_injury.csv *processed/weather/* weather.parquet *raw/weather/* stations.csv *raw/traffic/reference/* road_section_midpoints.csv, road_sections.parquet	*processed/* weather/road_period_frequency.csv traffic/road_period.csv	Builds road-period mean-wind exposure; road geometry is a fallback for missing midpoints.
traffic/rate_weather.py	*processed/* traffic/road_period.csv accidents/rural_injury.csv weather/weather.parquet *raw/weather/* stations.csv	*processed/accidents/* rate.csv	Aligns accident wind with the road-exposure station.
traffic/daily.py	*raw/traffic/daily_pdf/* *.pdf	*processed/traffic/* daily_raw.csv	Parses one daily count per counter channel and date.
traffic/download_roads.py	VGD-R MapServer layer 6	*raw/traffic/reference/* roads.geojson	Downloads the unchanged public road reference.
traffic/locate_counters.py	*processed/traffic/* daily_raw.csv *raw/traffic/reference/* roads.geojson	*processed/traffic/* daily.csv	Combines directional channels and locates counters.
traffic/daily_weather.py	*processed/* traffic/daily.csv weather/weather.parquet *raw/weather/* stations.csv	*processed/traffic/* daily_match.parquet daily_weather.csv	Matches counter-days to daytime and full-day wind; the compact Parquet join avoids a large CSV.
export_tables.py	*processed/* accidents/rural_injury.csv accidents/rate.csv accidents/case_control.csv weather/frequency.csv weather/traffic_frequency.csv traffic/annual.csv traffic/road_period.csv traffic/daily_weather.csv traffic/locations.csv	*analysis/* eleven data CSV files listed below	Selects only the variables used by ordinary analysis.

Table 2.4: Exact inputs and outputs of the ordinary analysis pipeline

Script	Input file(s)	Output file(s)	Description
tables/pipeline.py	docs/pipeline.md	*reports/thesis/* generated_pipeline_preparation.tex generated_pipeline_analysis.tex	Generates the two four-column thesis pipeline tables.
analysis/build_oe.py	analysis/accidents.csv analysis/accident_conditions.csv analysis/weather_frequency.csv	reports/working/tables/oe_station_bins.csv	Builds station-season observed and expected accident totals.
tables/oe.py	reports/working/tables/oe_station_bins.csv analysis/accidents.csv analysis/accident_conditions.csv	reports/main/tables/oe_results.csv mean_wind_oe.csv gust_oe.csv temperature_oe.csv weather_match_coverage.csv	Calculates clustered-bootstrap intervals and exports the retained O/E tables.
tables/radius_sensitivity.py	reports/main/tables/oe_results.csv	reports/main/tables/mean_wind_radius_sensitivity.csv	Selects the primary upper mean-wind estimates under 10, 20 and 30 km station limits.
figures/oe.py	reports/main/tables/oe_results.csv	reports/main/figures/mean_wind_oe.png gust_oe.png temperature_oe.png two mean-wind subgroup figures	Draws the O/E figures from the completed result table.
tables/annual_traffic_quality.py	analysis/annual_traffic.csv	reports/main/tables/annual_traffic_quality.csv	Audits nonpositive and unusual published seasonal traffic values.
tables/estimated_rate.py	analysis/traffic_exposure_full.csv	reports/main/tables/crash_rate_per_vehicle_km_by_wind.csv	Calculates the descriptive accident rate from all eligible road sections.
tables/rate.py	analysis/conditional_poisson_input.csv	*reports/main/tables/* conditional_poisson_rate_ratio_by_wind.csv conditional_poisson_rate_ratio_serious_fatal_by_wind.csv *reports/working/tables/* stratified_crash_rate_ratio_official_traffic.csv	Estimates injury and serious/fatal rate ratios and repeats the injury model using official VDU and SDU periods only.
figures/rate.py	The three tables written by tables/rate.py	*reports/main/figures/* conditional_poisson_rate_ratio_by_wind.png conditional_poisson_rate_ratio_serious_fatal_by_wind.png *reports/working/figures/* stratified_crash_rate_ratio_official_traffic.png	Draws the annual-traffic rate ratios.
tables/seasonal_rate.py	analysis/seasonal_poisson_input.csv	reports/main/tables/seasonal_poisson_rate_ratio_by_wind.csv	Fits four injury-accident models using coarse mean-wind intervals.
figures/seasonal_rate.py	reports/main/tables/seasonal_poisson_rate_ratio_by_wind.csv	reports/main/figures/seasonal_poisson_rate_ratio_by_wind.png	Draws four season-specific rate-ratio panels.
tables/daily_traffic.py	analysis/daily_traffic.csv	reports/main/tables/daily_traffic_by_wind.csv daily_traffic_period_summary.csv	Calculates traffic relative to the typical day at the same counter.

Table 2.4: Exact inputs and outputs of the ordinary analysis pipeline (continued)

Script	Input file(s)	Output file(s)	Description
`tables/daily_wind_duration.py`	`analysis/daily_traffic.csv`	`reports/main/tables/daily_traffic_by_high_wind_duration.csv`	Compares traffic with the calendar expectation by hours with `f >= 15 m/s`.
`figures/daily_wind_duration.py`	`reports/main/tables/daily_traffic_by_high_wind_duration.csv`	`reports/main/figures/daily_traffic_by_high_wind_duration.png`	Draws the sustained-wind traffic comparison.
`tables/daily_allocated_rate.py`	*analysis/* `accidents.csv` `accident_conditions.csv` `daily_traffic.csv` `daily_counter_locations.csv`	*reports/main/tables/* `daily_allocated_rate_ratio_by_wind.csv` `daily_allocated_rate_serious_fatal_by_wind.csv` `daily_allocated_rate_07_24_by_wind.csv`	Fits injury, serious/fatal, and 07:00–24:00 daily allocated-traffic models using accident-time `f`.
`figures/daily_allocated_rate.py`	`reports/main/tables/daily_allocated_rate_ratio_by_wind.csv`	`reports/main/figures/daily_allocated_rate_ratio_by_wind.png`	Draws the retained daily-counter injury-accident rate ratios.
`tables/daily_counter_rate.py`	*analysis/* `accidents.csv` `daily_traffic.csv` `daily_counter_locations.csv`	`reports/main/tables/daily_counter_rate_ratio_by_wind.csv` `daily_counter_rate_ratio_coarse_by_wind.csv`	Fits the detailed and coarse full-day observed-traffic comparisons within counter and year.
`tables/daily_counter_radius.py`	*analysis/* `accidents.csv` `daily_traffic.csv` `daily_counter_locations.csv`	`reports/main/tables/daily_counter_radius_sensitivity.csv`	Repeats both non-reference coarse estimates at 5, 10 and 20 km.
`figures/daily_counter_rate.py`	`reports/main/tables/daily_counter_rate_ratio_coarse_by_wind.csv`	`reports/main/figures/daily_counter_rate_ratio_coarse_by_wind.png`	Draws the coarse observed-traffic comparison with 95% intervals.
`tables/traffic_sensitivity.py`	Main and official-period rate tables `analysis/daily_traffic.csv` `reports/main/tables/annual_traffic_quality.csv`	`reports/main/tables/traffic_sensitivity.csv`	Places the alternative traffic-period rule, zero-counter-day check, and annual-traffic quality checks in one table.
`figures/daily_traffic.py`	`reports/main/tables/daily_traffic_by_wind.csv`	`reports/main/figures/daily_traffic_by_wind.png`	Draws the supporting daily-traffic result.
`figures/data_flow.py`	`analysis/selection_summary.csv` `reports/main/tables/weather_cleaning_audit.csv`	*reports/main/figures/* `accident_flow.png` `weather_flow.png` `traffic_flow.png`	Draws accident, weather and traffic selection figures.
`figures/accident_profiles.py`	`analysis/accidents.csv`	*reports/main/figures/* `accident_types.png` `vehicles_per_accident.png` `accident_types_by_severity.png`	Draws accident-type, vehicle-count and severity descriptions.
`tables/conditions.py`	`analysis/accidents.csv` `analysis/accident_conditions.csv`	`reports/main/tables/accident_conditions_summary.csv` `temperature_coverage.csv`	Summarises hour, traffic period, daylight and temperature.
`figures/conditions.py`	`reports/main/tables/accident_conditions_summary.csv`	`reports/main/figures/accident_conditions_overview.png`	Draws the consolidated raw-count description of accident conditions.
`tables/case_control.py`	`analysis/case_control.csv`	`reports/main/tables/case_control_weather.csv`	Fits continuous and categorical conditional logistic models.
`tables/high_wind_profile.py`	`analysis/accidents.csv` `analysis/accident_conditions.csv`	`reports/main/tables/high_wind_accident_profile.csv`	Compares the primary sample with accidents at mean wind at least 15 m/s.
`validate.py`	Analysis CSVs and retained result tables	`reports/main/tables/final_analysis_validation.md`	Checks row counts, included records, and headline results.

Urban, and valid points outside all polygons as Rural. A defined Vestmannaeyjar area is also classified as Urban. The same 2020–2024 polygon dataset is applied to the full 2007–2025 accident period. Changes in urban boundaries before 2020 are therefore not represented.

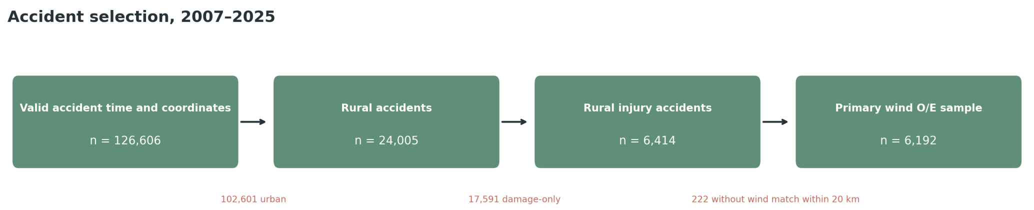

Figure 2.1: Available accident records, severity of the rural injury sample, and coverage of the primary weather match. Damage-only accidents remain available but are not part of the injury outcome.

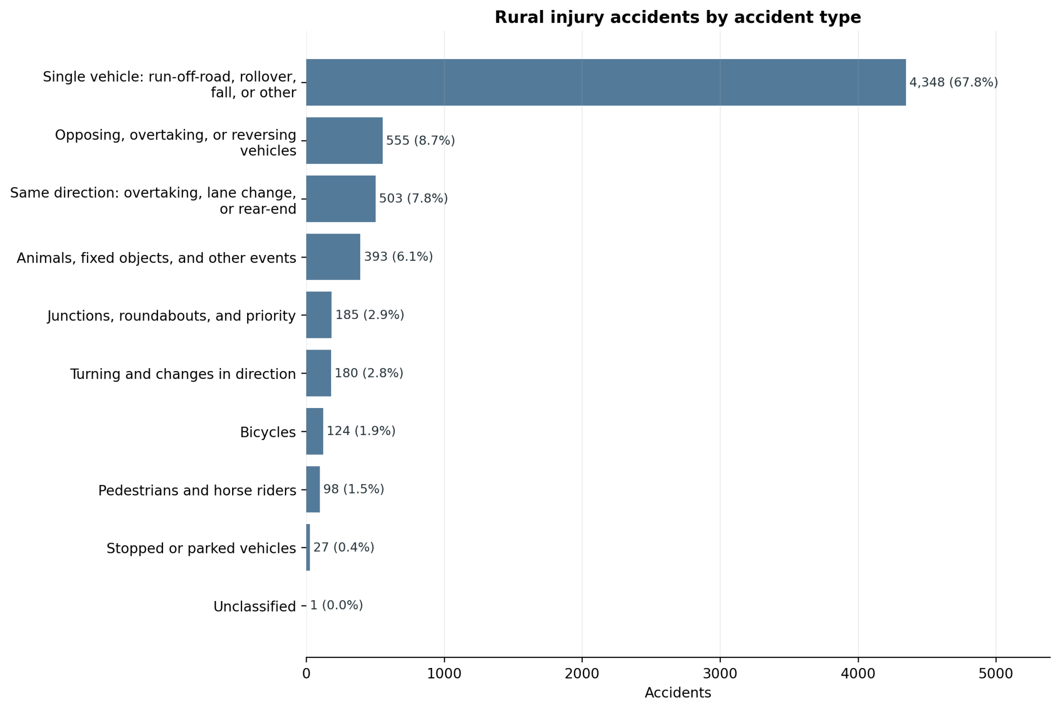

Figure 2.2: Accident-type distribution among the 6,414 rural injury accidents.

2.3. Weather Data

The weather source contains 10-minute observations from Icelandic weather stations. The variables used here are mean wind speed (`f`), reported wind gust (`fg`), and temperature (`t`). For each accident, the gust is taken from the same matched ten-minute observation as mean wind; it is not the maximum across the accident day. Mean wind remains the primary exposure. A wind observation is used when

$$0 \leq f < 45, \qquad 0 \leq fg < 65, \qquad fg + 0.5 \geq f.$$

Rows missing either `f` or `fg` are excluded. Negative values, values at or above either upper limit, and internally inconsistent values where `fg = 0` but `f > 0` are excluded and reported in the annual audit. A value of `f = 0` alone is retained. Only uninterrupted all-zero runs where `f = fg = 0` for at least two hours are excluded as potentially frozen sensors. Missing temperature does not affect whether a wind observation is used.

Each accident is matched to the geographically nearest station with a clean wind observation at either adjacent timestamp on the 10-minute observation grid. The observations therefore have 10-minute resolution, but the selected observation is no more than five minutes from the recorded accident time. The primary maximum distance is 20 km. The matching file retains all 6,414 accidents and records whether each accident meets the weather-match criterion.

Temperature is matched independently because the nearest station with valid wind does not always report temperature. The nearest temperature observation between -30 and 30°C within the same 20 km and five-minute limits is selected. The temperature station, distance, and time difference are retained separately. Wind and temperature may therefore come from different stations. This increases usable temperature coverage without changing the wind sample or discarding accidents with missing temperature.

Table 2.5: Wind-quality audit of all delivered station-time rows, 2007–2025

Category	Records	Share of total
All delivered station-time rows	226,580,952	100.00%
Rows from station-years without wind data	11,810,743	5.21%
Records assessed for wind quality	214,770,209	94.79%
Missing `f` or `fg`	427,778	0.19%
Negative or upper-threshold wind	300,649	0.13%
Internally inconsistent `f`/`fg`	2,268	<0.01%
Frozen all-zero runs ($\geq$ 2 hours)	2,541,617	1.12%
Clean wind observations retained	211,497,897	93.34%

Thus, 93.34% of all delivered rows are retained. Among the 214,770,209 rows from station-years that contain wind measurements, retention is 98.48%. The detailed audit reports each cleaning rule separately. The audit and the weather-distribution figure use the full 2007–2025 weather delivery. For each accident's station and season, the O/E denominator uses all clean background observations in that station–season group, not only observations recorded on accident days.

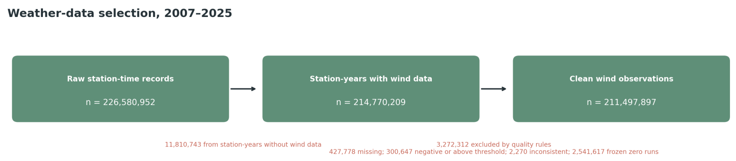

Figure 2.3: Weather observations retained after the documented wind-quality rules.

2.4. Traffic Data

Two traffic sources are used for different purposes.

Annual road-section files provide average annual daily traffic (ADU), summer daily traffic (SDU), winter daily traffic (VDU), section length, and annual vehicle-kilometres for 2000–2025 [Vegagerðin(2026)]. The road-section comparison uses the years shared with the accident study, 2007–2025. Following the official definitions, SDU covers June–September and VDU covers December–March. These files can be linked to accidents with a registered road-section code, but exact matching is much narrower than the weather matching.

Table 2.6: Traffic periods used in the annual-traffic analysis.

Period	Months	Daily traffic reference
VDU	December–March	Official winter daily traffic
SDU	June–September	Official summer daily traffic
VHDU	April–May and October–November	Derived from ADU after the VDU and SDU totals have been removed

Thus, spring is April–May and autumn is October–November in the traffic analysis. VHDU is a transparent residual estimate, rather than an additional official traffic measure.

Daily traffic counts extracted from Vegagerðin reports cover 2019–2024. They describe observed traffic at counter sites and are useful for measuring travel demand during windy days. They do not provide hourly vehicle-kilometres at every accident location. More than one counter may also occur on a road section. Each report identifies a counter by road section and *stöð* (road station). The station value is placed along the official road geometry and marked as an estimated location; annual-file start and end stations are used only as a fallback where the official station range is unavailable. They are therefore used only as a supplementary description of whether travel demand changes on windy days; they are not part of the primary O/E denominator.

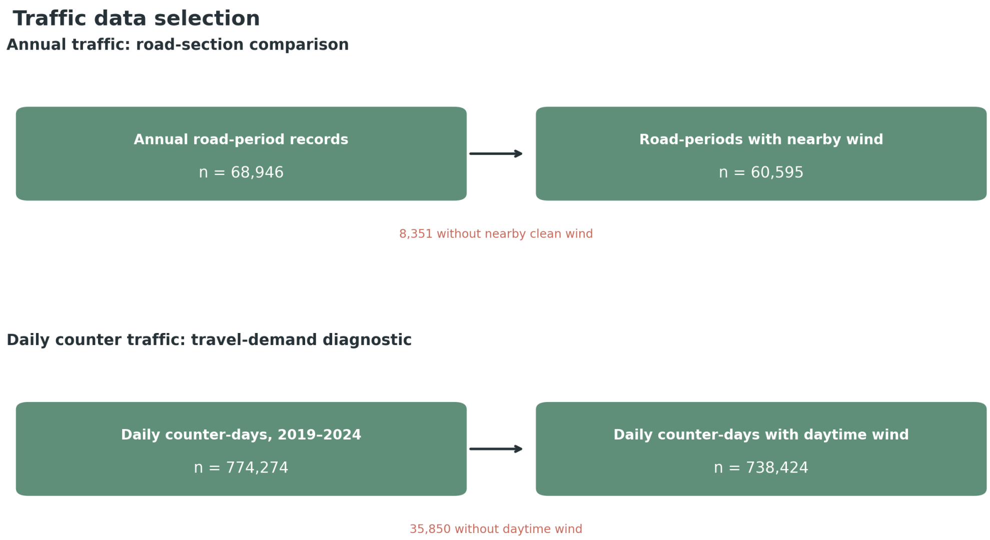

Figure 2.4: Coverage of the annual road-section data, the road-section wind table, and the separate daily-counter diagnostic.

2.5. Reproducible Implementation

The repository groups the code into weather, accident, traffic, table, and figure modules. Preparation scripts clean and match source records, table scripts calculate numerical results, and figure scripts draw completed tables. The repository README identifies the individual programs. After the authorised source deliveries have been placed locally, the reported results can be regenerated with

```
.venv/bin/python -m src.prepare --stage prepare
.venv/bin/python -m src.analyze
```

Figures are generated by the figure scripts and are not manually edited.

3. Methods

3.1. Primary Wind-Frequency Analysis

Mean wind speed `f` is the pre-specified primary weather measure and is divided into 5 m/s intervals. Wind gust `fg` from the ten-minute observation matched to the accident time is retained as a supporting analysis. Each accident is compared with the mean-wind distribution at its matched weather station in the same meteorological season, pooled across all study years. The expected count in a mean-wind interval is therefore based on how often that interval occurred locally, rather than on a national wind average.

Temperature uses the same station-and-season O/E structure with independently matched temperature stations. The intervals are below -5, -5 to -3, -3 to -1, -1 to 1, 1 to 3, 3 to 5, and at least 5°C. These narrow intervals around freezing follow the exploratory temperature specification; the open tails retain every valid match.

Let g identify a weather station and meteorological season. Let A_{jg} be the number of accidents in mean-wind interval j and group g, and let $A_g = \sum_j A_{jg}$. From all clean 10-minute observations, let T_{jg} be the number in interval j and T_g the total in group g. The local background frequency is

$$p_{jg} = \frac{T_{jg}}{T_g}.$$

If accident timing followed local wind frequency, the expected number in the interval would be $A_g p_{jg}$. The pooled observed/expected ratio is

$$R_j = \frac{\sum_g A_{jg}}{\sum_g A_g p_{jg}}.$$

A value of one means that the accident share equals the local time share of the mean-wind interval. A value above one means that accidents are over-represented in that interval. Standardisation within station and season prevents windy stations or seasons from dominating solely because their wind distributions differ.

Uncertainty is estimated with 5,000 weather-station-clustered bootstrap replicates. Stations are sampled with replacement, while all seasons, weather observations, and accidents associated with a sampled station remain together. The reported interval is the 2.5th to 97.5th percentile of the bootstrap distribution.

3.2. Time-Stratified Case-Crossover Analysis

As an independent comparison, each accident time is matched to the other occurrences of the same weekday and clock time in the same calendar month and year. Wind controls use the accident's matched wind station; temperature controls use its independently matched temperature station. Control weather must satisfy the same five-minute observation-time rule, and both station matches are restricted to 20 km. Conditional logistic regression compares weather at the accident time with weather at its matched control times. This time-stratified referent strategy was selected to avoid overlap bias and to control calendar time without selecting controls from outside the accident month [Janes et al.(2005)].

This design controls exactly for weather station, year, month, weekday, and clock time. It therefore controls recurring travel schedules more tightly than the seasonal O/E analysis without estimating hourly traffic from daily counts. It does not control unusual traffic changes, road conditions, or other time-varying causes on a particular day. Continuous effects are reported per 5 m/s or 5°C, together with categorical estimates. This analysis is supporting evidence; the pre-specified mean-wind O/E remains primary.

3.3. Why Traffic Is Not in the Primary Ratio

The primary denominator is background wind time. Traffic is not included because it is unavailable at the spatial and temporal resolution required for the full study period. The estimated crash-rate analysis uses annual traffic on every eligible road section, rather than only sections where an accident occurred. It includes 4,958 accidents that have an exact annual road-section, year, and traffic-period link. It uses SDU and VDU where they are published, and a clearly labelled residual VHDU estimate for April–May and October–November.

Using those traffic data in the primary curve would change the study population and mix seasonal average traffic with 10-minute accident-time wind. The primary curve therefore measures *wind-frequency-adjusted relative accident occurrence*. It does not itself measure accident rate per vehicle.

3.4. Road-Section Traffic Analysis

Traffic is included where the spatial and temporal keys can be linked explicitly. For road section, year, official traffic period, and wind interval k, j, estimated vehicle-kilometres are

$$V_{kj} = q_k L_k D_k p_{kj},$$

where q_k is SDU, VDU, or derived VHDU vehicles per day, L_k is section length, D_k is the number of calendar days, and p_{kj} is the local share of clean 10-minute observations in the wind interval. The accident rate is

$$\text{Rate}_j = \frac{A_j}{\sum_k V_{kj}} \times 100{,}000{,}000.$$

This method accounts for road length and for spatial and seasonal differences in published traffic. It still assumes that traffic within a traffic period is distributed across wind intervals in proportion to time. It does not observe traffic at the accident minute. The accident wind is measured at the nearest qualifying station to the accident location; the wind frequency used to allocate traffic is measured at the nearest qualifying station to the road-section midpoint. For this analysis, the accident wind is re-matched to that same station and retained only when the station is also within 20 km of the accident.

The principal traffic result is a conditional Poisson model. Each road section, year, and traffic period has its own intercept. For stratum k and wind interval j, the model is

$$\log E(A_{kj}) = \alpha_k + \beta_j + \log V_{kj},$$

where A_{kj} is the accident count and $\log V_{kj}$ is the offset. The exponentiated coefficient $\exp(\beta_j)$ is the estimated rate ratio relative to 0–5 m/s within otherwise identical road-section–year–period strata.

The model is fitted separately for all injury accidents and for serious or fatal accidents. A seasonal extension uses year-specific station frequencies for winter, spring, summer, and autumn. Because individual 5 m/s upper bins are sparse within seasons, that extension uses 0–10, 10–15, and at least 15 m/s. Winter and summer use published VDU and SDU, respectively; spring and autumn use the same derived VHDU daily volume but retain separate weather frequencies and accident counts.

The descriptive rate and conditional model use different exposure universes. The descriptive rate sums vehicle-kilometres over every eligible road section. The conditional model retains only road-section–year–period strata containing at least one accident, because all-zero strata provide no within-stratum information. Positive-exposure wind-bin rows are retained within those strata. The two traffic results therefore answer different questions and are not expected to reproduce the same rate ratios.

The annual source contains 22,982 road-section/year rows for 2007–2025. There are 1,509 rows with nonpositive published VDU and 552 rows with a nonpositive derived VHDU residual. Exposure for the affected period is excluded rather than imputed. Seasonal patterns such as SDU below VDU are audited but are not automatically treated as errors.

The daily counter analysis is kept separate from the vehicle-kilometre model. It describes whether traffic at selected counters is lower on windier days; it does not directly observe vehicle-kilometres in each 10-minute wind interval.

3.5. Observed Daily Traffic Diagnostic

Daily traffic is analysed separately because its weather definition is daily. Directional channels are first summed to one 24-hour count for each physical counter and date. The unit is then one fixed counter on one date; separate counter sites on the same road section are not summed. For each counter, year, month, and weekday, expected traffic is the arithmetic mean of the available daily counts in that same group. The traffic index is

$$I_{cd} = 100 \times \frac{Q_{cd}}{\mathrm{mean}(Q_{c,\,\text{same year, month, weekday}})}.$$

An index below 100 indicates fewer counted vehicles than the mean for that local calendar context. For each wind interval, the observed-to-expected traffic ratio is aggregated across counter-days; 100 denotes the calendar-standardised expected traffic. The arithmetic mean is used so expected and observed totals balance within a complete calendar stratum; medians are retained only in separate counter-quality summaries. It describes the participating counter sites rather than national traffic. The 24-hour traffic count is not converted into hourly traffic. Mean wind is calculated from clean observations between 10:00 and 21:59, so this is a daily association rather than an estimate of traffic in that window. The source contains 5,432 counter-days with a reported count of zero. They are retained because the reports do not distinguish a genuine zero or road closure from counter downtime. Consequently, the daily result is supplementary and the result with zero counts excluded should be inspected before interpretation. The report's *stöð* is placed proportionally between the official start and end stations on the registered road-section geometry. The resulting coordinate is explicitly marked as estimated rather than treated as a surveyed counter position. This locates 2,325 of 2,332 counter-site years; the seven unresolved site-years remain without a weather match. An independent check finds an official 20 m road-station point within 10 station metres for 435 of 473 distinct located sites; among those matches, the median coordinate difference is 5.0 m and the 90th percentile is 9.0 m. Failure to find a point within that narrow query window is retained as unresolved rather than silently accepted. For each located counter, the nearest station within 20 km having usable observations that day is selected. Uncertainty in the daily traffic estimate is obtained by resampling whole counters 1,000 times.

Counter-year mean traffic is compared with exact road-section ADU as a consistency check. A near-complete counter-year has at least 300 observed days; single-counter road sections are evaluated separately because multiple PDF records can represent different locations, directions, or lanes.

Sustained wind is described by the number of valid ten-minute observations in each counter-day with mean wind of at least 15 m/s; six observations equal one hour. Days must contain at least 108 valid observations. Traffic is compared across 0, more than 0–2, 2–6, and at least 6 hours of strong wind using the same counter–calendar standardisation.

The allocated daily-counter rate model uses a stricter accident linkage. Each 2019–2024 accident requires a counter on the exact registered road section and year within 20 km, positive traffic on its date, and qualifying accident-time mean wind. For counter (c), date (d), and wind interval (j), the observed 24-hour count is allocated according to the counter station's valid ten-minute observations:

$$\widehat{Q}_{cdj} = Q_{cd}\frac{N_{cdj}}{N_{cd}}.$$

Accidents are classified by their matched accident-time `f`. Exposure and accidents are aggregated by counter, year, and wind interval, and a conditional Poisson model gives each counter–year its own intercept. The allocation assumes uniform traffic across valid observation times; it is estimated within-day exposure, not observed hourly traffic. Accident and counter weather stations are selected independently and can differ. The full-day-mean model is retained only as an appendix day-level check. The main allocation uses all sufficiently complete 24-hour weather records. A separate check assumes all daily traffic occurs from 07:00 to midnight, uses only weather observations and accidents in that interval, and is reported only to show the effect of that unverified assumption. Both versions are also fitted for the serious-or-fatal outcome where counts permit.

4. Results

4.1. Coverage

Table 4.1: Coverage of the retained analyses

Analysis step	Available	Data included	Coverage
Wind match within 10 km	4,858	6,414	75.74%
Wind match within 20 km (primary)	6,192	6,414	96.54%
Wind match within 30 km	6,399	6,414	99.77%
Temperature match within 20 km	5,707	6,414	88.98%
Exact annual road-section match, 2007–2025	5,731	6,414	89.35%
Stratified crash-rate analysis	4,958	6,192	80.07%
Daily counter-days with daytime wind	738,424	774,274	95.37%
Allocated daily-counter rate, 2019–2024	760	1,863	40.79%

The 20 km threshold retains most accidents while limiting the spatial distance to the weather proxy. Increasing the threshold to 30 km adds 207 accidents but can be misleading in complex terrain. The road-section analysis is substantially narrower than the primary analysis and must not be presented as equally representative.

The two annual-traffic rows have different analytical meanings. The first records an exact road-section and year link. The second also requires a defined traffic period and usable annual traffic and weather-frequency exposure. It is the population used for the estimated crash-rate analysis below.

The selection chain is therefore explicit: 126,607 source accidents are reduced to 6,414 rural injury accidents by the outcome and location rules; 6,192 of these have a qualifying primary wind match; 5,731 have an exact road-section/year traffic link; and 4,958 enter the conditional traffic model. Within 2019–2024, 767 of 1,863 accidents have the stricter exact-section, distance, and valid observed counter-day link used by the direct daily-rate check. Each reduction answers a different data requirement and no unmatched accident is silently reassigned.

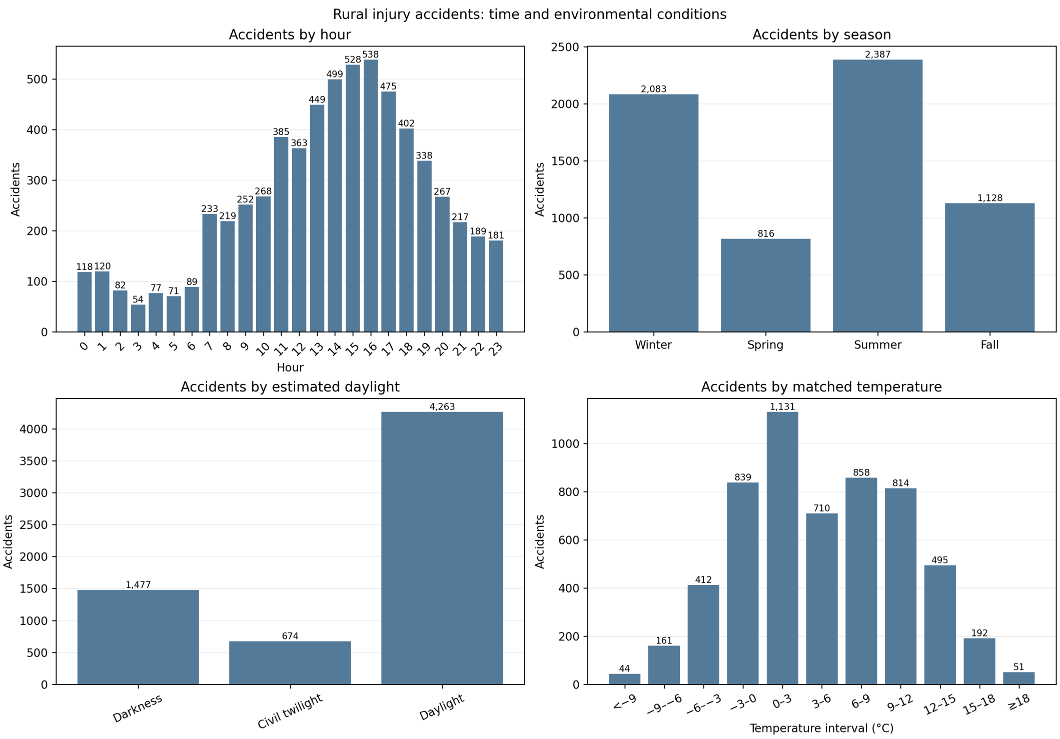

Figure 4.1: Raw counts of rural injury accidents by recorded hour, season, estimated daylight, and matched temperature. Hour, season, and daylight use all 6,414 accidents; temperature uses the 5,707 accidents with a qualifying temperature match. These panels describe the study sample and do not estimate risk because traffic exposure differs between categories.

4.2. Mean Wind Speed and Relative Accident Frequency

Ratios are close to one across most low and moderate intervals. The plotted ratios increase from 15 m/s. At mean wind speeds of 20–25 m/s, 61 accidents were observed compared with 29.0 expected. The O/E ratio is 2.11, with a station-clustered interval of 1.54–2.73. At mean wind speeds of at least 25 m/s, 16 accidents were observed compared with 6.4 expected (O/E 2.49; 95% interval 0.86–5.37). This interval includes one and is too imprecise to support a separate conclusion at 25 m/s or above. The more informative upper mean-wind result is the 20–25 m/s interval.

The 5 m/s grouping is used throughout because it gives a direct, readable description of the upper wind intervals without implying that the boundaries are physical thresholds.

The upper-wind pattern is stable under the pre-defined spatial alternatives (Table 4.2). At 20–25 m/s, O/E is 2.32 using 10 km, 2.11 using the primary 20 km limit, and 2.05 using 30 km. The wider radii add accidents but do not create the association. The ≥ 25 m/s interval remains imprecise under every radius because it contains only 15–17 accidents.

Table 4.2: Primary mean-wind O/E under three weather-station distance limits

Radius	Primary	Mean wind	Observed	Expected	O/E (95% interval)
10 km	No	15–20	192	123.8	1.55 (1.34–1.76)
		20–25	53	22.8	2.32 (1.67–3.06)
		≥ 25	15	5.3	2.85 (0.90–6.20)
20 km	Yes	15–20	223	157.1	1.42 (1.23–1.60)
		20–25	61	29.0	2.11 (1.54–2.73)
		≥ 25	16	6.4	2.49 (0.86–5.37)
30 km	No	15–20	227	162.9	1.39 (1.20–1.58)
		20–25	62	30.2	2.05 (1.51–2.68)
		≥ 25	17	6.9	2.47 (0.93–5.23)

Temperature remains exploratory. Valid temperature was available for 5,707 accidents (88.98%). Coverage by year ranges from 78.86% to 97.37%, so no temperature value is imputed. After standardisation by temperature station and season, the largest departure was in the 3–5°C interval (464 observed, 691.2 expected; O/E 0.67, station-bootstrap 95% CI 0.61–0.74). The -1 to 1°C interval was elevated (769 observed, 609.0 expected; O/E 1.26, 95% CI 1.17–1.35), whereas the open warm tail was close to the seasonal expectation (O/E 1.01). The non-monotonic pattern may reflect travel behaviour, precipitation, or road conditions and is not interpreted as a causal temperature effect or as an additional thesis conclusion.

4.3. Comparison with Matched Non-accident Times

The case-crossover analysis retains 6,190 wind strata with 20,869 control times and 5,697 temperature strata with 19,175 control times. A 5 m/s increase in mean wind is associated with a matched odds ratio of 1.08 (95% CI 1.04–1.12). The categorical estimate changes principally at mean wind of at least 15 m/s: the odds ratio relative to 0–5 m/s is 1.61 (95% CI 1.39–1.87). This supports the high-wind pattern after exact control for station, year, month, weekday, and clock time.

The temperature estimates are not monotonic. In particular, both -5–0 and at least 15°C are elevated relative to 0–5°C, while the continuous estimate decreases. This differs from a simple interpretation of the warm-temperature O/E result and reinforces its exploratory status.

Among the 6,192 accidents in the primary wind sample, 300 occurred at mean wind of at least 15 m/s. Winter accounts for 51.7% of these high-wind accidents, compared with 32.8% of the primary sample. Single-vehicle accidents account for 63.0% versus 72.1%, and serious or fatal accidents for 20.0% versus 22.7%. These are descriptive compositions, not adjusted risk comparisons.

4.4. Results Using Traffic Data

The conditional Poisson model includes 4,958 accidents in 4,004 road-section–year–traffic-period groups that contain accidents. Relative to 0–5 m/s, the estimated within-stratum rate ratio is 1.59 (95% CI 1.36–1.86) at 15–20 m/s, 2.38 (1.78–3.18) at 20–25 m/s, and 4.49 (2.75–7.34) at at least 25 m/s. The uppermost estimate contains 17 accidents and is therefore less precise. These rates use traffic allocated within each period according to local wind frequency; they are not directly observed 10-minute vehicle exposure.

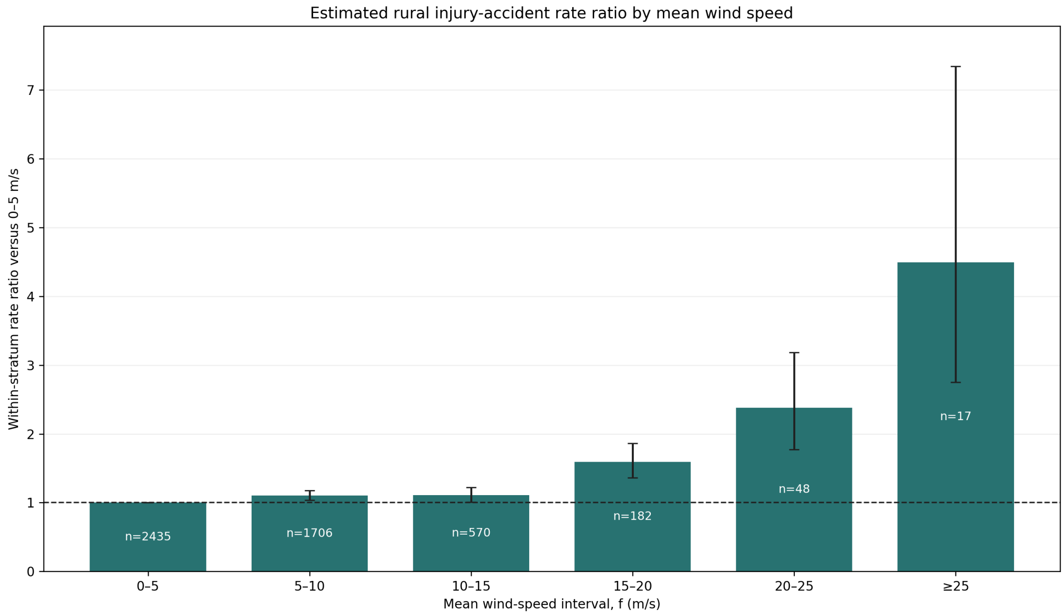

Figure 4.3: Estimated rural injury-accident rate ratios relative to 0–5 m/s. The model compares wind intervals within the same road section, year, and traffic period. Traffic is allocated within a period in proportion to local 10-minute wind frequency. Labels show observed accidents and error bars are 95% confidence intervals.

The serious-or-fatal model retains 1,054 accidents. Relative to 0–5 m/s, its rate ratio is 1.90 (95% CI 1.38–2.60) at 15–20 m/s and 2.17 (1.11–4.25) at 20–25 m/s. The at-least-25 m/s estimate is 4.70 (1.69–13.08), but contains only four accidents and is correspondingly imprecise.

Using coarse seasonal intervals, the rate ratio for at least 15 versus 0–10 m/s is 1.45 (95% CI 1.19–1.76) in winter, 2.85 (1.98–4.11) in spring, 2.12 (1.55–2.90) in summer, and 1.38 (1.02–1.87) in autumn. The corresponding upper-category counts are 122, 34, 44, and 47. The pattern is therefore not confined to winter, although spring and autumn use derived VHDU traffic and should not be compared as precise seasonal rankings.

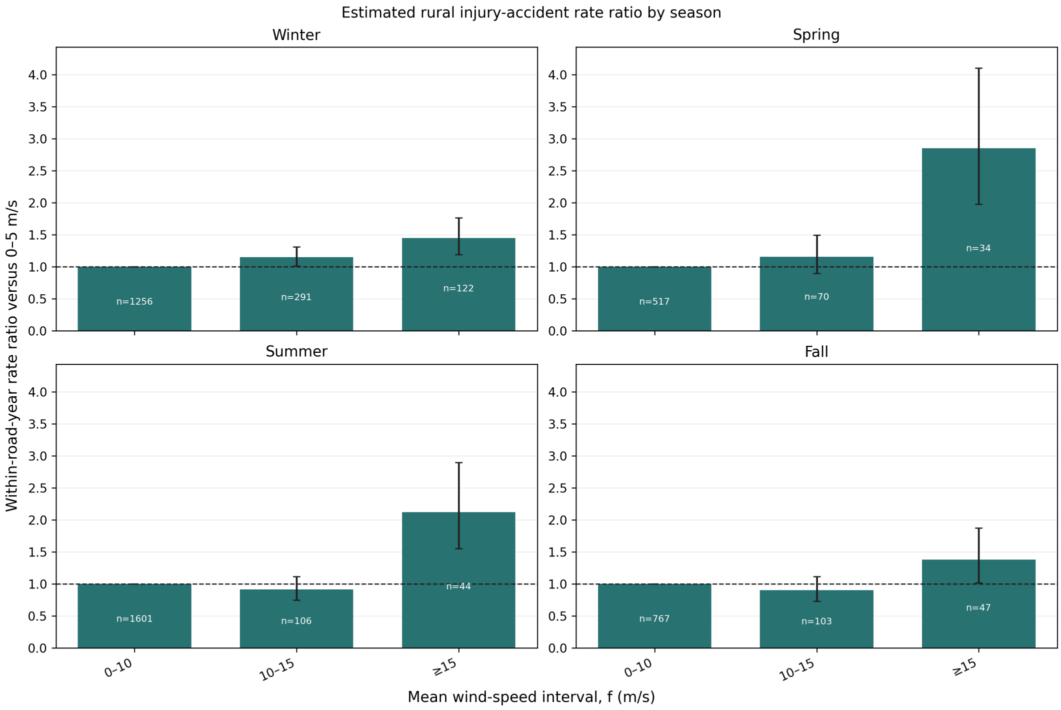

Figure 4.4: Estimated rural injury-accident rate ratios by season. Traffic and weather frequency are aligned by road section, year, and season. Coarse wind intervals avoid unstable one-event upper bins; labels show observed accidents and error bars are 95% confidence intervals.

Observed traffic decreases as strong wind persists. Relative to the calendar expectation at the same counter, traffic is 96.4% on days with more than zero but less than two hours at $f \geq 15$ m/s, 94.4% with 2–6 hours, and 88.9% with at least six hours. This duration measure is more informative than a single daily mean, but still describes selected counter sites.

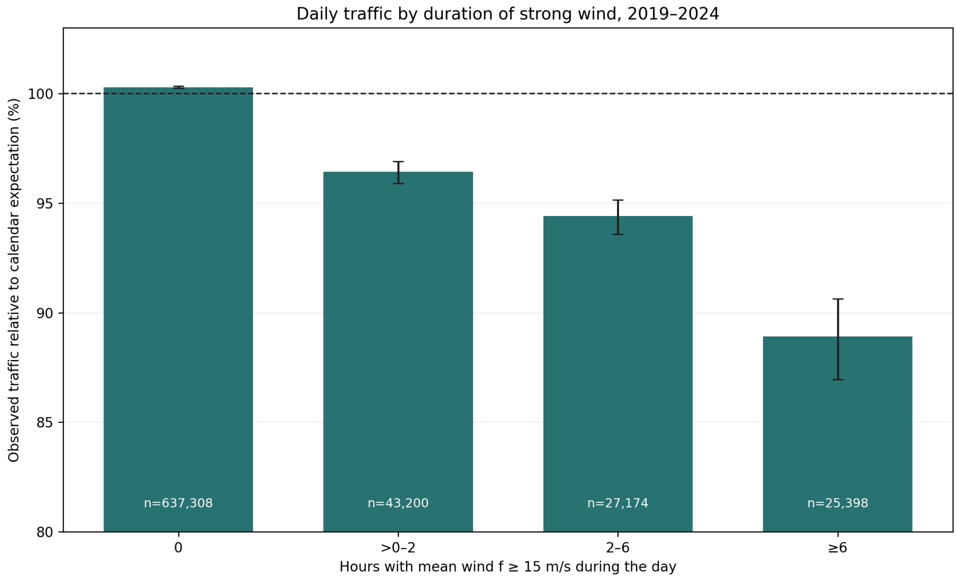

Figure 4.5: Observed daily traffic by duration of strong wind. Values are relative to the mean for the same counter, year, month, and weekday. Error bars are counter-clustered 95% confidence intervals; labels show counter-days.

The allocated daily-counter model retains 760 accidents in 488 counter–year groups that contain accidents. Relative to 0–10 m/s, the rate ratio is 1.77 (95% CI 1.44–2.17) at 10–15 m/s and 3.38 (2.46–4.62) for at least 15 m/s. The latter category contains 46 accidents. The denominator starts from observed daily traffic, but its allocation within the day is estimated from the counter station's ten-minute wind frequency; it is not observed hourly traffic. For serious or fatal accidents, the corresponding estimates are 1.82 (1.17–2.82; 25 accidents) at 10–15 m/s and 4.16 (2.25–7.69; 12 accidents) for at least 15 m/s. Restricting the allocation and accident times to 07:00–24:00 changes the all-injury upper estimate only modestly, from 3.38 (2.46–4.62; 46 accidents) to 3.50 (2.53–4.83; 44 accidents). This stability does not validate the assumed time distribution; it only shows that the headline direction is not created by including midnight–07:00 weather.

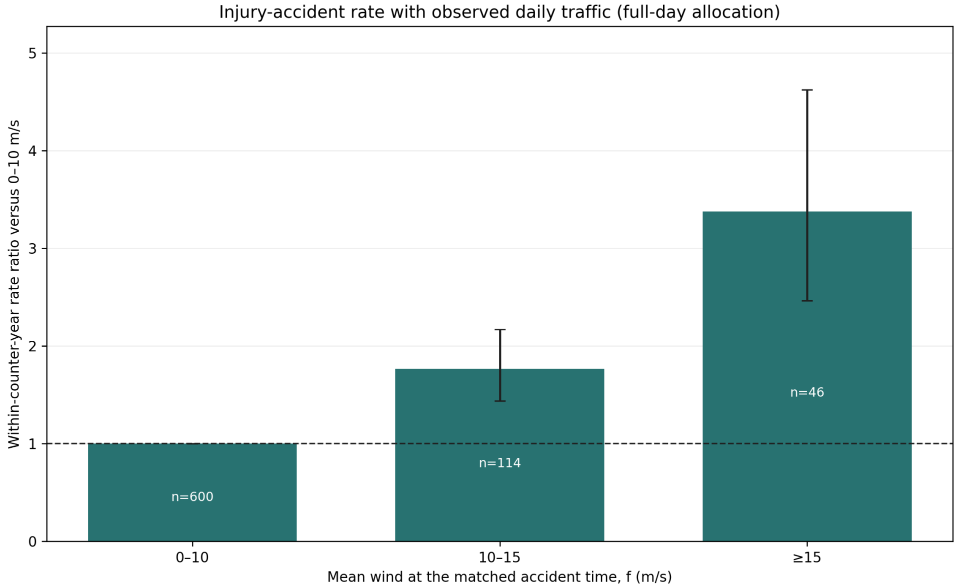

Figure 4.6: Daily-counter rural injury-accident rate ratios by accident-time mean wind. Observed 24-hour traffic is allocated within each day in proportion to the counter station's ten-minute wind frequency. Comparisons are within counter and year; labels show accidents and error bars are 95% confidence intervals.

Table 4.3 addresses the two most consequential data questions. Restricting the rate model to the official VDU and SDU periods reduces the upper estimates but retains the same direction. Excluding daily counter records reported as zero changes the 20–25 m/s traffic estimate from 74.9% to 75.2%, so those records do not explain the observed decline.

Table 4.3: Compact comparison and quality checks for the traffic analyses

Check	Data included	Estimate (95% interval)	Records
Rate model, 20–25 m/s	All periods, including derived VHDU	RR 2.38 (1.78–3.18)	48 accidents
	Official VDU and SDU only	RR 1.93 (1.32–2.82)	28 accidents
Rate model, ≥ 25 m/s	All periods, including derived VHDU	RR 4.49 (2.75–7.34)	17 accidents
	Official VDU and SDU only	RR 2.52 (1.18–5.35)	7 accidents
Daily traffic, 20–25 m/s	Retain zero counter-days	74.9% (67.2–80.5)	1,359 days
	Exclude zero counter-days	75.2% (67.8–80.8)	1,275 days
Annual-traffic quality	Nonpositive published VDU	6.57% excluded	1,509 section-years
	Nonpositive derived VHDU	2.40% excluded	552 section-years

Table 4.4: Summary of the evidence used to interpret the main result

Analysis	Comparison	Estimate (95% interval)	Interpretation and principal limitation
Primary O/E	20–25 m/s	2.11 (1.54–2.73)	Accident occurrence relative to local wind frequency; traffic is not included.
Case-crossover	$\geq$ 15 vs 0–5 m/s	OR 1.61 (1.39–1.87)	Exact calendar-time matching; no direct control for unusual daily travel changes.
Conditional Poisson	20–25 vs 0–5 m/s	RR 2.38 (1.78–3.18)	Within road, year, and traffic period; vehicle-kilometres are estimated from annual traffic.
Allocated daily rate	$\geq$ 15 vs 0–10 m/s	RR 3.38 (2.46–4.62)	Observed daily total with estimated within-day allocation; 760 linked accidents, including 46 in the upper category.

The four rows estimate different quantities and should not be compared by magnitude. Their value is the consistent interpretation: accidents are over-represented in strong wind even though observed traffic tends to be lower on windy days. The separate daily traffic-volume diagnostic is retained in the appendix because it describes travel response rather than accident occurrence.

Matched-time wind gust is a secondary measure. Its O/E pattern rises more strongly in the highest gust intervals; at gusts of at least 35 m/s, 28 accidents were observed compared with 5.5 expected (O/E 5.06; 95% interval 2.94–7.57). This result is reported as supporting evidence, not as a replacement for the mean-wind primary analysis.

Table 4.5: Highest-gust result under three weather-station distance rules

Maximum distance	Matched accidents	Observed at $\geq$ 35	O/E	95% interval
10 km	4,858	28	6.12	3.58–9.26
20 km (primary)	6,192	28	5.06	2.94–7.57
30 km	6,399	29	5.00	2.95–7.45

4.5. Matched-Time Wind Gust as a Supporting Analysis

Matched-time wind gust is examined with the same station and season standardisation. The gust O/E figure is provided in Appendix A.2.

4.6. Descriptive Subgroups

Appendix figures repeat the mean-wind calculation by season and by the number of vehicles involved. They are descriptive checks, not separate research questions or tests of interaction; the high-wind cells become sparse after the sample is divided. Hour and daylight are presented only as accident counts in the appendix. A valid O/E subgroup requires the same classification for accident times and for every background weather observation. Daylight has not yet been derived for the complete station-time denominator, and the available road- surface history describes road construction rather than the wet, icy, or dry condition at the accident time. Presenting either comparison now would be misleading.

The traffic data cannot supply the missing hourly denominator. Annual traffic is seasonal, while the counter data contain one total for an entire day. The 07:00–24:00 uniform allocation is therefore used only as an assumption check and not as an estimated hourly traffic profile. No traffic-adjusted hour or daylight O/E curve is reported.

The appendix is organised by purpose. Accident-characteristic figures describe the selected study population. The gust result and mean-wind subgroup figures check whether the main pattern is confined to one exposure definition or subgroup. Detailed daily and annual traffic tables document the supporting traffic work. None is used to redefine the primary outcome after seeing the results.

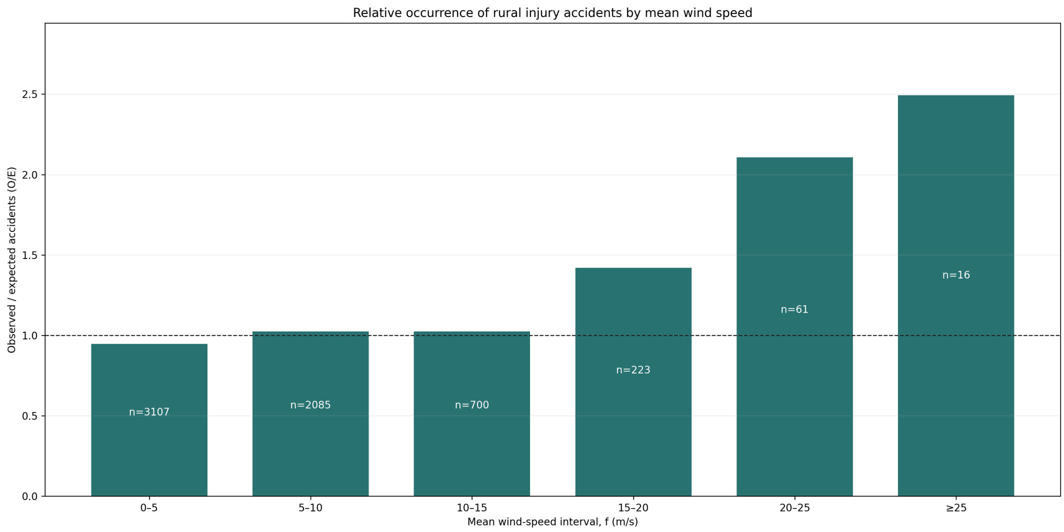

Figure 4.2: Wind-frequency-adjusted relative occurrence of rural injury accidents by mean wind speed. The label inside each bar is the observed accident count. Traffic volume is not included. Bootstrap intervals are reported in the supporting results rather than drawn on the figure.

5. Discussion

5.1. Interpretation

Strong mean wind is over-represented at the times of rural injury accidents relative to its local frequency. The comparison is standardised within weather station and season, and uncertainty is clustered by station. The result describes association rather than a per-vehicle causal effect. Moderate wind intervals fluctuate around one, while several upper intervals contain few accidents.

Traffic remains important for interpretation. The primary curve cannot separate risk per vehicle from the number of vehicles exposed. The principal traffic-model result is therefore reported alongside the primary result, with detailed diagnostics in the appendix. The traffic analyses indicate lower traffic during strong wind, but their temporal and spatial coverage is not sufficient to make them a final national estimate of accident risk per vehicle-kilometre.

The daily-counter analyses support the same interpretation. Observed traffic is lower when strong wind persists for more hours, and the allocated daily model again gives rate ratios above one. That model starts from measured 24-hour traffic but estimates its within-day distribution from wind frequency. It covers 760 accidents near selected counters and is therefore supporting rather than nationally representative evidence.

The serious-or-fatal and four-season models point in the same direction as the all-injury result. They broaden the evidence without changing the primary outcome or exposure definition. Sparse extreme-wind counts, especially the four serious-or-fatal accidents in the at-least-25 m/s interval, preclude precise comparison of effect sizes across outcomes or seasons.

The O/E and traffic-model estimates are not competing versions of the same number. O/E compares accident occurrence with the amount of time each wind interval is observed, using 6,192 matched accidents. The conditional model compares estimated accident rates within road section, year, and traffic period and uses 4,958 accidents with the required traffic links and positive exposure. Its larger upper-bin estimate is compatible with traffic falling during strong wind, but it also depends on allocated rather than directly observed 10-minute traffic. The agreement in direction is more informative than the difference in magnitude.

5.2. Strengths

The main strengths are the large 2007–2025 accident sample, 10-minute weather resolution, explicit spatial coverage reporting, local wind-frequency standardisation, and station-clustered bootstrap intervals. A separately specified time-stratified analysis controls station and recurring calendar time, while the traffic analyses examine the primary denominator limitation. The matching data retain unmatched accidents and record the reason for each unsuccessful match. The fixed hierarchy keeps the connection between the research question, method, and result clear.

5.3. Limitations

- The primary result is not traffic-adjusted. It should not be interpreted as accident risk per vehicle-kilometre.
- The case-crossover analysis controls recurring calendar time but not unusual traffic, road conditions, visibility, or closures on a particular day. Its ≥ 15 m/s category is a supporting summary of the upper-wind pattern rather than a physical threshold.
- A weather station up to 20 km away is a proxy for conditions at the accident location. Terrain can produce important differences over shorter distances.
- The stratified crash-rate analysis represents 4,958 accidents and uses annual seasonal traffic allocated by wind frequency, rather than traffic observed at the accident minute. It aligns accident weather and exposure to the same station, but excludes accidents for which that station is farther than 20 km from the accident.
- Daily counter data cover 2019–2024. The duration analysis relates the observed 24-hour total to hours of strong wind; it does not observe traffic separately during those hours.
- Daily traffic counts cover selected counter sites. Multiple counters can represent different directions, lanes, or locations; retaining counters separately avoids double counting but does not provide national coverage.
- The allocated daily-rate model retains 760 accidents with an exact road-section counter link and valid counter-day. It assumes the observed daily traffic is distributed across wind intervals in proportion to valid ten-minute observations. This is an estimated within-day split, not observed hourly traffic; the accident and counter weather stations can also differ, and the accident vehicle need not have passed the assigned counter.
- The 07:00–24:00 daily check assumes zero traffic outside that interval and uniform traffic within it. This is deliberately an assumption check, not a

reconstruction of hourly traffic.

- Estimated rates are restricted to rural injury accidents on linked road sections. They should not be compared directly with broad national or international accident rates unless the outcome and vehicle-kilometre denominator are defined in the same way.
- The highest wind intervals contain relatively few accidents. Several intervals and comparison results are shown, so individual estimates in the upper tail should be interpreted with caution.
- Fixed 2020–2024 urban boundaries are applied to the entire 2007–2025 accident period.
- The analysis is observational. Unmeasured road condition, vehicle type, speed, visibility, and driver behaviour can contribute to the pattern.

5.4. Next Improvements

The most valuable additional data would be verified hourly traffic counts with counter coordinates, stable road-section identifiers, lane and direction metadata, and documented downtime. Linking those counts to the same weather stations would provide a genuine hourly denominator for accident and control times. Daily totals cannot recover that denominator, so further modelling of the present totals would not solve this limitation. With hourly data, a future analysis could distinguish vehicle classes, wind direction relative to the road, and exposed-road geometry. Better exposure data would add more value than additional models based on the current information.

6. Conclusion

This thesis examines wind speed and rural injury accidents in Iceland. Of 6,414 accidents from 2007–2025, 6,192 were matched to a valid wind observation within 20 km. After standardisation for local mean-wind frequency within station and season, accidents were over-represented at mean wind speeds of 20–25 m/s: 61 were observed compared with 29.0 expected, giving an O/E ratio of 2.11 with a 95% station-clustered interval of 1.54–2.73. The secondary gust analysis shows a stronger upper-tail pattern.

The time-stratified and traffic-based results support the same direction as the primary O/E analysis. However, neither traffic source gives complete hourly exposure for the full study period. The principal conclusion is therefore an association between strong mean wind and higher relative occurrence of rural injury accidents. Better linked hourly traffic data are required to estimate directly observed road risk per vehicle.

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A. Supporting Analyses

These figures are retained because each documents the study population, checks the primary pattern using another comparison, or examines the traffic data. They are descriptive unless stated otherwise; sparse bars are shown in grey and their observed accident counts are printed on the bars.

A.1. Accident Characteristics

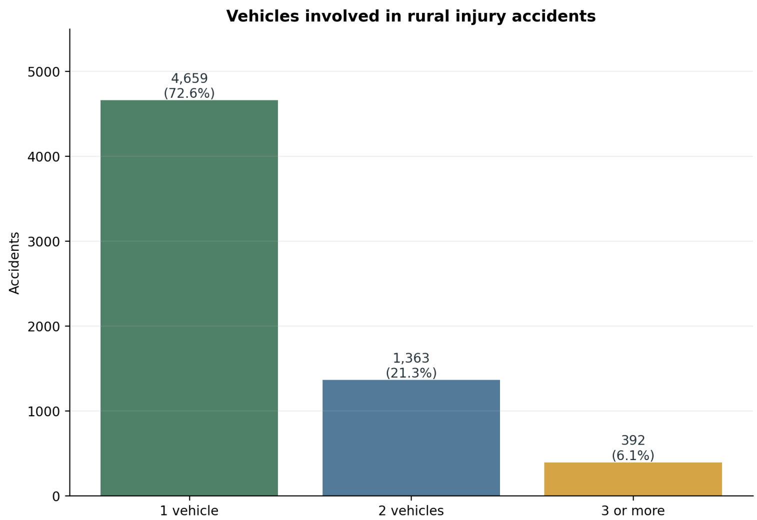

Figure A.1: Number of registered vehicles involved in the 6,414 rural injury accidents. Most selected accidents involved one registered vehicle.

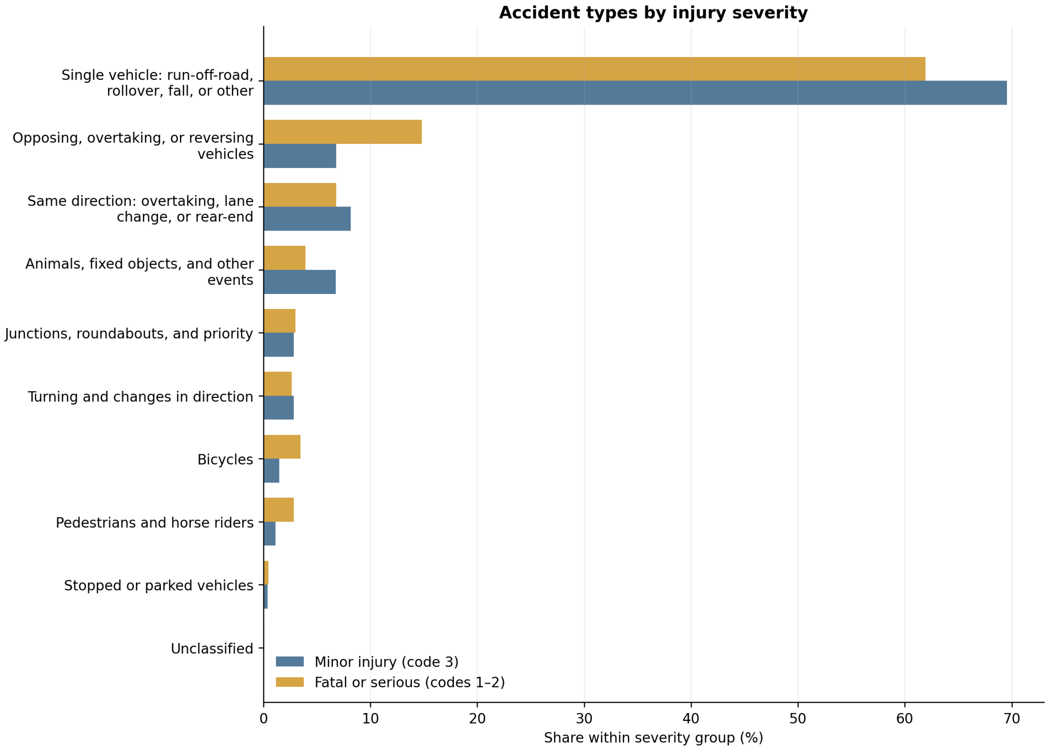

Figure A.2: Accident-type composition in minor-injury accidents and in the non-overlapping fatal-or-serious group. Percentages sum to 100 within each severity group.

A.2. Wind Gust at the Matched Accident Time

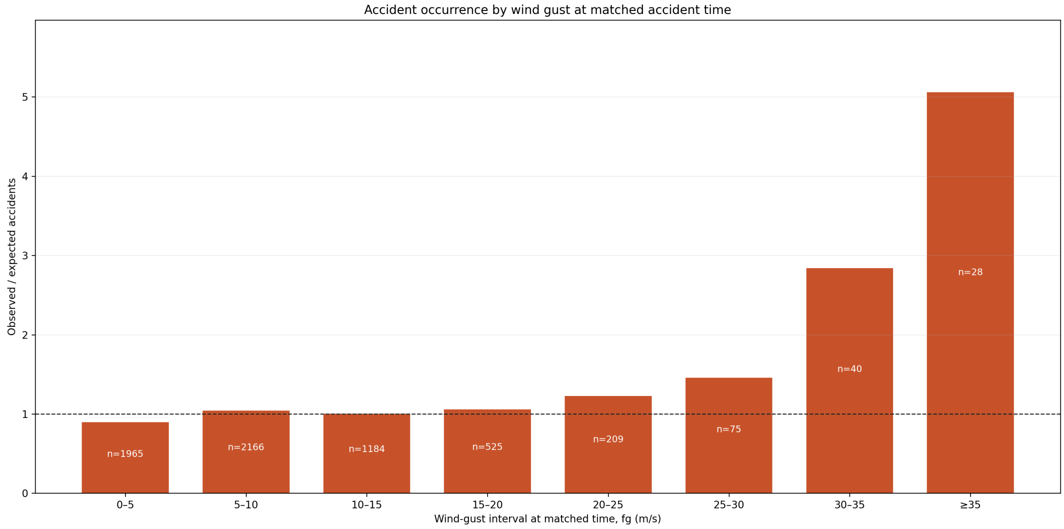

Figure A.3: Frequency-adjusted observed/expected accident occurrence by wind gust in the observation matched to the accident time. The upper-tail increase supports, but does not replace, the primary mean-wind result. Labels show observed accidents.

A.3. Mean-Wind Subgroups

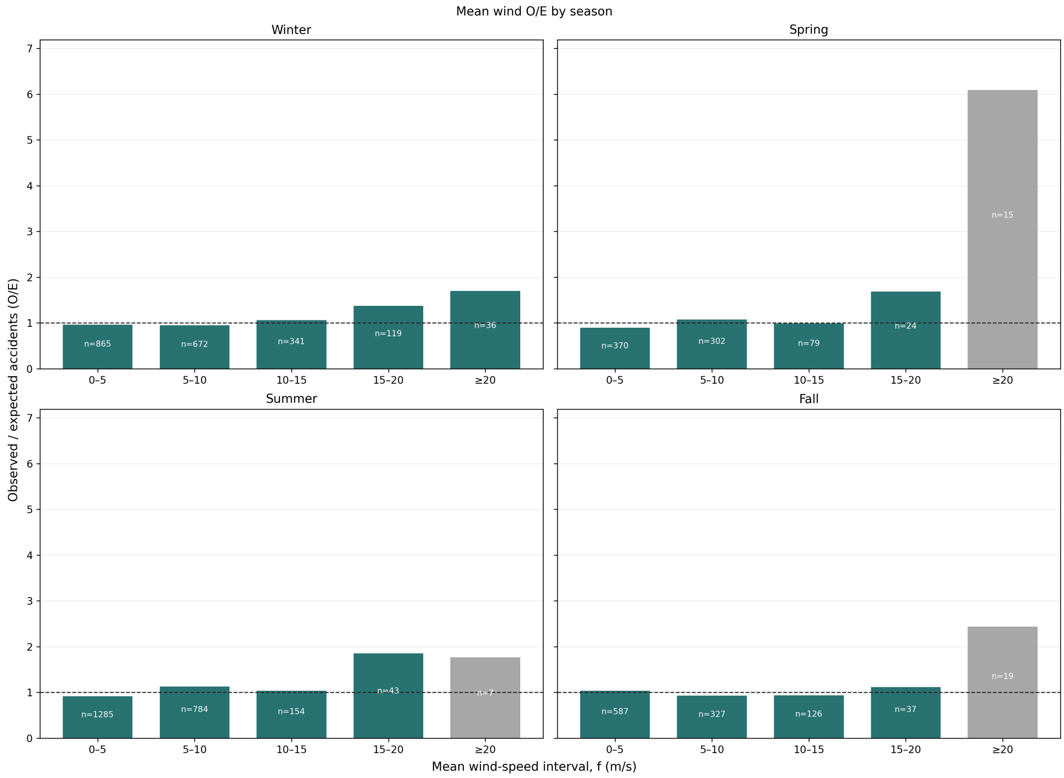

Figure A.4: Mean-wind O/E estimates by meteorological season. These estimates are shown as a descriptive subgroup analysis; high-wind bins are sparse after the sample is divided by season. All panels share one vertical scale; grey bars contain fewer than 20 accidents.

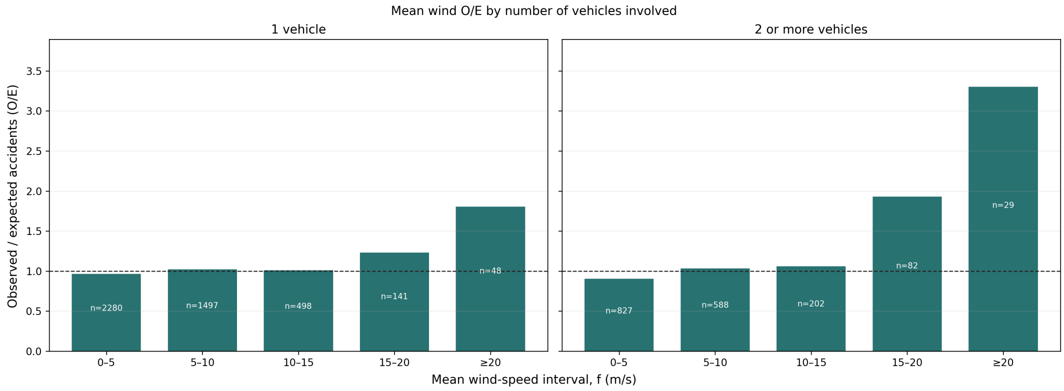

Figure A.5: Mean-wind O/E estimates by the number of registered vehicles involved. This is a descriptive subgroup analysis; it does not establish accident type or vehicle count as an explanatory variable. Both panels share one vertical scale; grey bars contain fewer than 20 accidents.

A.4. Temperature, Hour, and Daylight

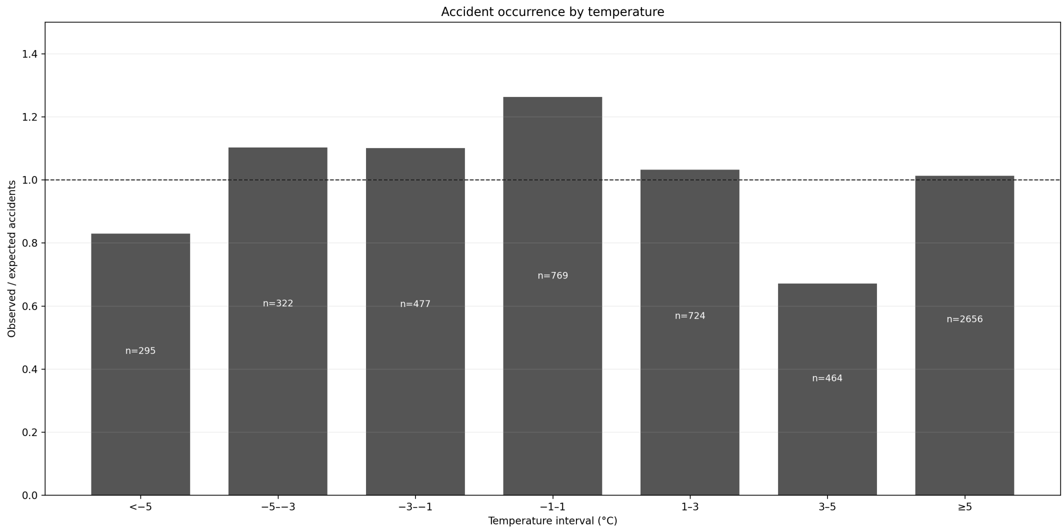

Figure A.6: Temperature O/E estimates standardised by the independently matched temperature station and meteorological season. Temperature is exploratory; mean wind remains the primary weather measure. Labels give observed accident counts.

The temperature bars compare observed accidents with the number expected from the local seasonal temperature distribution. The intervals are below -5, -5 to -3, -3 to -1, -1 to 1, 1 to 3, 3 to 5, and at least 5°C. The open tails ensure that no valid observation is omitted. Raw temperature counts document the sample but cannot indicate relative occurrence because temperatures have very different background frequencies.

The raw-count panels in Figure 4.1 show that the largest hourly counts are at 15:00 and 16:00. Overall, 66.5% of the selected accidents occurred in daylight, 10.5% in civil twilight, and 23.0% in darkness. These values describe when the study accidents occurred; they do not show that daylight or a particular hour is safer or more hazardous. Such a comparison would require traffic counts by hour and daylight condition, which is not available in the daily-counter data.

A.5. Daily-Counter Validation

Table A.1: Counter-location construction and independent coordinate check

Check	Result
Counter-site years in analysis location table	2,332
Counter-site years located from road geometry	2,325
Distinct located counter sites	473
Sites matched to an official 20 m road-station point within 10 m	435
Median / 90th-percentile coordinate difference	5.0 / 9.0 m

Coordinates are interpolated from the reported road station along official road geometry and remain labelled as estimates. The separate 20 m point match tests the interpolation; it is not used to overwrite the primary location.

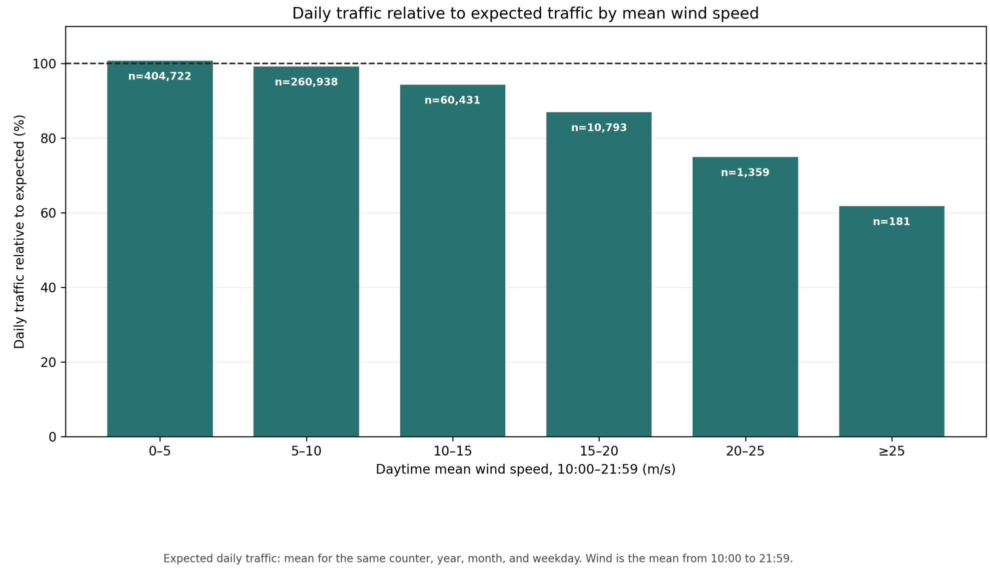

Figure A.7: Vehicles counted per day relative to a typical day at the same counter, year, month, and weekday. Labels show counter-days. Daytime mean wind is calculated from 10:00 to 21:59. This is a supporting description of selected counter sites, not traffic across the national road network.

Table A.2: Consistency of observed daily PDF traffic with official ADU

Data included	N	Median	P10–P90	$r_{\log}$
All exact matches	2,328	0.975	0.245–1.115	0.942
At least 300 observed days	1,906	0.984	0.240–1.063	0.940
At least 300 days, one counter per section	1,122	1.000	0.925–1.112	0.993

A.6. Full-Day-Mean Daily-Counter Check

Table A.3: Selection of accidents for the observed daily-counter analysis

Selection step	Accidents	Share of 2019–2024 sample
Rural injury accidents in 2019–2024	1,863	100.0%
Exact road-section/year counter candidate	864	46.4%
Assigned counter within 20 km	863	46.3%
Positive count and valid full-day wind	767	41.2%

The main loss occurs because many accident road sections have no daily counter record for the same year. Distance excludes only one additional candidate at 20 km; a further 96 accidents lack a positive counter count or valid full-day wind on the accident date. No accident is assigned across road sections merely to improve coverage.

Table A.4: Detailed daily counter-based rural injury-accident rate comparison, 2019–2024

Mean wind	Accidents	Counted vehicles	Counters	Rate ratio (95% CI)
0–5 m/s	392	284,393,198	230	1.00 (reference)
5–10 m/s	273	164,363,094	230	1.05 (0.90–1.24)
10–15 m/s	88	29,868,420	230	1.71 (1.34–2.18)
15–20 m/s	12	2,719,300	176	2.43 (1.35–4.37)
20–25 m/s	1	80,251	64	4.32 (0.59–31.37)
$\geq$ 25 m/s	1	3,578	6	170.95 (18.19–1,606.66)

The detailed table documents the standard wind intervals but is not used for upper-tail inference: the final two estimates are determined by one accident each and are extremely unstable. The coarse version combines the upper tail and gives a rate ratio of 2.42 (95% CI 1.38–4.24) for at least 15 m/s. Both versions use the same 767 accidents, exact road-section/year assignment within 20 km, and observed 24-hour traffic as the offset. Because a full-day mean does not represent conditions at the accident time, this analysis is retained only as a day-level check.

Table A.5: Upper daily-counter estimate under three assignment distances

Maximum distance	Included accidents	Upper-bin accidents	Rate ratio	95% CI
5 km	585	12	2.65	1.44–4.88
10 km	750	14	2.48	1.41–4.35
20 km	767	14	2.42	1.38–4.24

The estimate remains in the same direction under stricter distance rules. The wide intervals reflect the small number of accidents at full-day mean wind of at least 15 m/s; stability across radii does not remove that limitation.

A.7. Stratified Vehicle-Kilometre Model

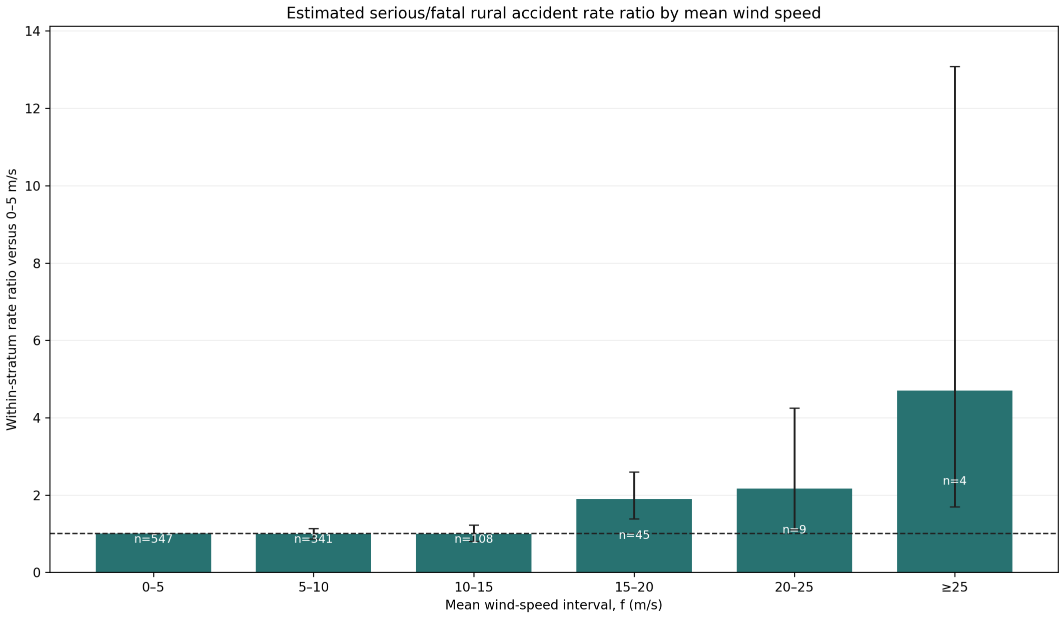

Figure A.8: Estimated serious-or-fatal rural accident rate ratios by mean wind. The model compares wind intervals within the same road section, year, and traffic period. Labels show observed accidents. The four accidents in the highest interval make that estimate imprecise.

Table A.6 gives descriptive absolute rates using estimated vehicle-kilometres from every eligible road section. Table A.7 compares the all-period conditional model with a model restricted to official VDU and SDU traffic periods. The latter excludes April–May and October–November, for which VHDU is derived from ADU, SDU and VDU.

Table A.6: Estimated rural injury-accident rate by mean wind speed, all traffic periods

Mean wind	Accidents	Estimated accidents per 100 million vehicle-km
0–5	2,435	8.5
5–10	1,706	13.2
10–15	570	16.4
15–20	182	27.3
20–25	48	43.0
≥ 25	17	57.2

Table A.7: Conditional rate ratios in the upper mean-wind intervals: all periods versus official VDU and SDU periods only

Mean wind	All periods	Official VDU+SDU only
15–20	1.59 (1.36–1.86), $n = 182$	1.67 (1.39–2.01), $n = 131$
20–25	2.38 (1.78–3.18), $n = 48$	1.93 (1.32–2.82), $n = 28$
≥ 25	4.49 (2.75–7.34), $n = 17$	2.52 (1.18–5.35), $n = 7$

The two models agree that the rate ratio is higher from 15 m/s. The upper intervals differ because the official-only comparison excludes VHDU and has few accidents. This comparison is retained to show the dependence of the estimated rate on the traffic-period assumption.

The four-season rate model is reported in the main results using wider wind intervals to avoid very small cells. A vehicle-group rate model is not retained because the upper intervals become too sparse for a balanced comparison. The descriptive O/E vehicle-group figure above shows the limited evidence available.